SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer**.** Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer**.** Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT)**.** Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server

and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

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RUBRICS FOR CALCULATION

Criteria	**Weight ag e**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understand in g of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technol ogi es	Good understandi ng with proper integration of most concepts	Basic understandi ng ; limited or partial integration	Poor understandi ng ; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementat i on with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Document ati on	10	Clear, well structured documentation with diagrams, code, and results	Organized documenta tio n with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n

3.ARP Request and ARP Reply Using Cisco Packet Tracer

AIM

To construct a LAN with multiple hosts connected through a switch, generate ARP Request and ARP Reply packets using ping, and analyze how ARP maps IP addresses to MAC addresses.

TOOLS USED

- Cisco Packet Tracer
- Wireshark
- 3 PCs
- 1 Switch
- Ethernet cables

PROCEDURE

1. Open **Cisco Packet Tracer**.
2. Place **3 PCs** and **1 Switch** in the workspace.
3. Connect all PCs to the switch using **Copper Straight-Through** cables.
4. Configure the IP addresses:
    - PC0 → 192.168.1.10
    - PC1 → 192.168.1.20
    - PC2 → 192.168.1.30
    - Subnet Mask → 255.255.255.0
5. Open PC0 → **Command Prompt** and check the ARP table using:

arp -a

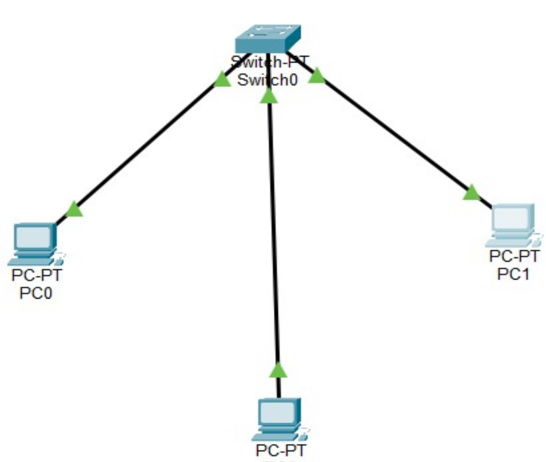
6. Switch to **Simulation Mode** in Packet Tracer.
7. Select **ARP** and **ICMP** in the filters.
8. From PC0, execute:

ping 192.168.1.20

9. Use **Capture/Forward** to observe the packets.
10. Identify the **ARP Request** and **ARP Reply**.
11. Examine the sender and target IP and MAC addresses.
12. Observe the ICMP Echo Request and Echo Reply after ARP resolution.

OUTPUTS:

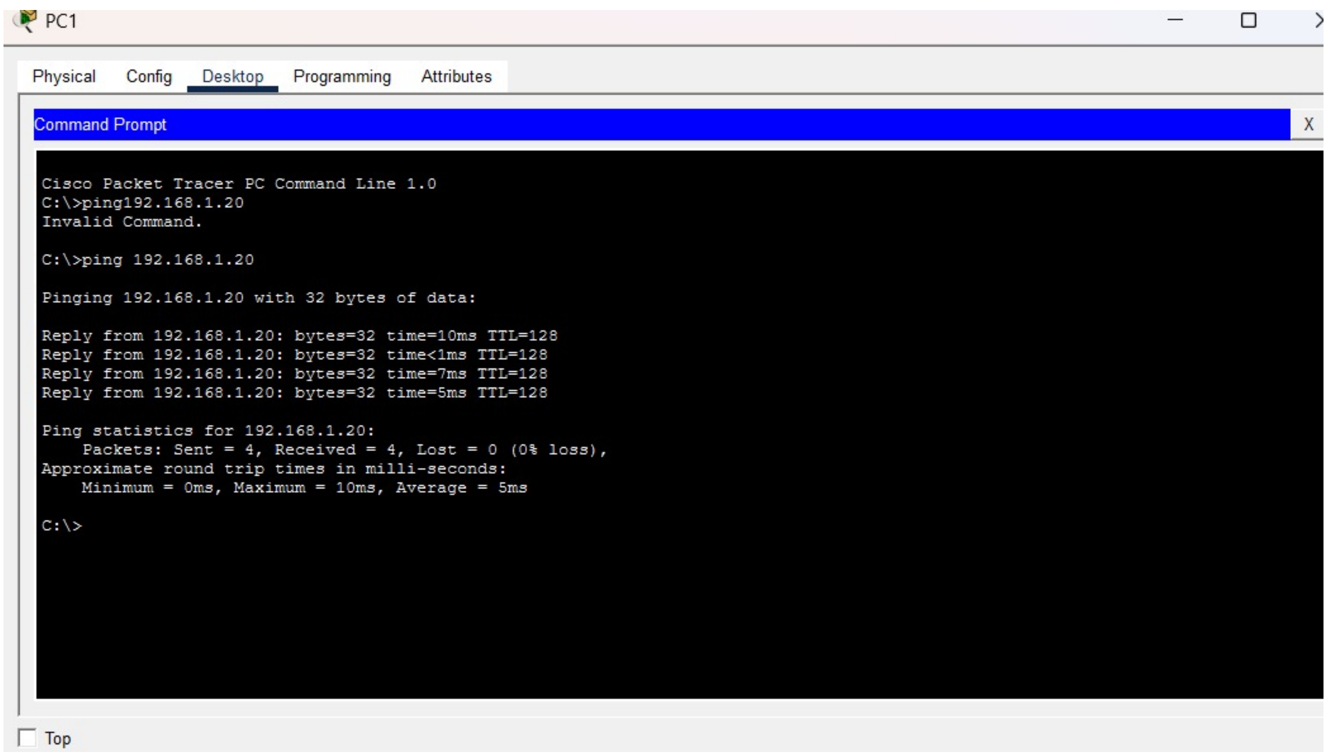
Construct a local area network with multiple hosts connected through a switch in Cisco



Packet Tracer

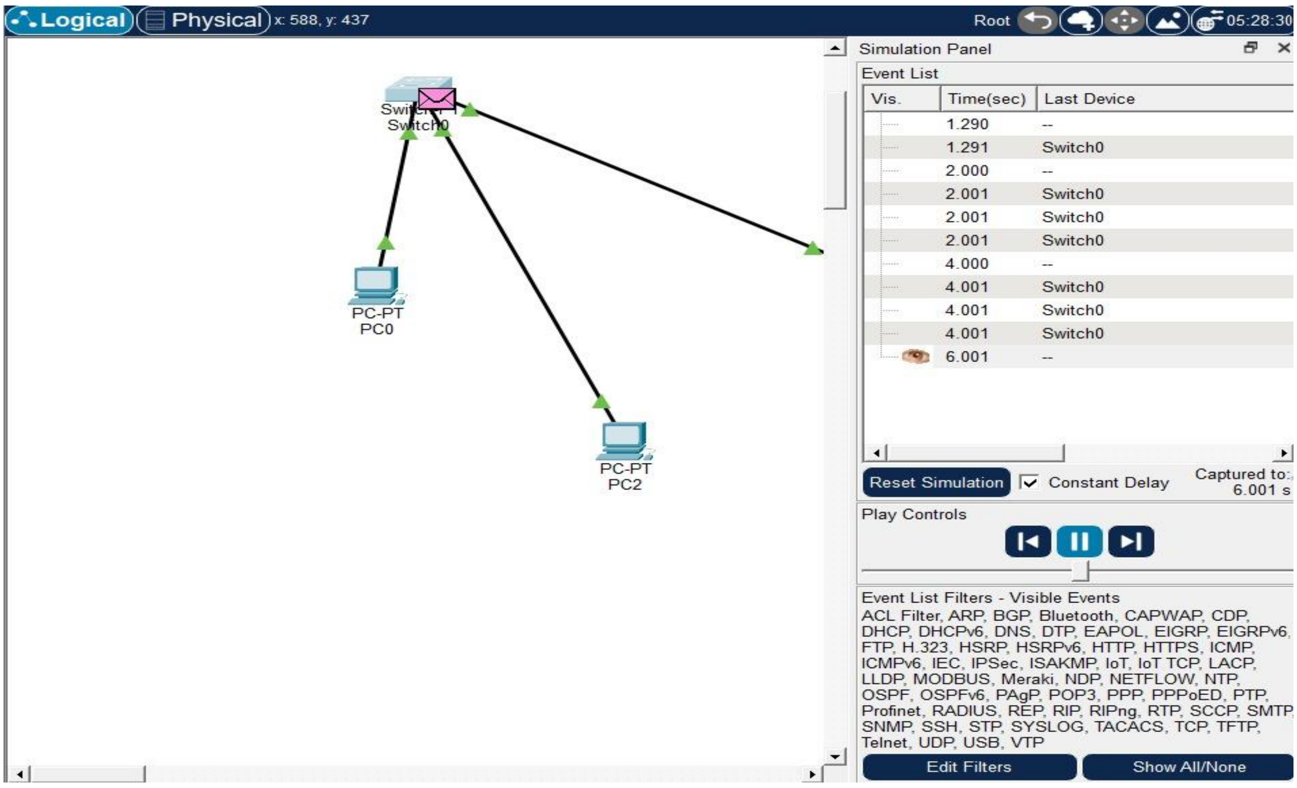
- **Network Topology:** Three PCs (**PC0, PC1, and PC2**) are connected to a central **Switch0**, forming a simple Local Area Network (LAN).
- **Communication:** The switch provides communication between all three PCs, allowing them to exchange data using their assigned **IP and MAC addresses**.

2.Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply
packets.



- Ping Test: PC1 successfully sent ping packets to 192.168.1.20.
- Result: 4 packets were sent and 4 were received with 0% packet loss, confirming successful connectivity.

<u>**3.**</u> Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.



- Simulation: The packets are successfully transmitted through Switch0 between the connected PCs.
- Result: The Simulation Panel displays the packet events, confirming successful communication through the LAN.

RESULT

Thus, the LAN was successfully created and the **ARP Request and ARP Reply** were observed. ARP successfully resolved the destination **IP address into its corresponding MAC address** before data .